OFFICIAL PROJECT TECHNICAL REPORT & SPECIFICATION

AUDIT V PLATFORM

Enterprise Financial Intelligence, Autonomous Tax Compliance Engine & AI Advisory System

SYSTEM VERSION	PRIMARY FRAMEWORK	AI / ML CORE	STORAGE ENGINES
v1.0.0 (Tax Year 2026-27)	FastAPI + React 18 + Vite	Qwen 2.5 3B (QLoRA) + Isolation Forest	SQLite ORM + ChromaDB HNSW

1. Executive Summary & Project Overview

Audit V is an enterprise-grade financial intelligence, tax calculation, and fraud risk detection platform designed specifically for the Indian taxation, regulatory, and corporate financial ecosystem. The platform seamlessly bridges *deterministic statutory computation* (which demands 100% mathematical precision without LLM hallucination) and *probabilistic artificial intelligence* (which enables natural language question answering, contextual guidance, voice navigation, and pattern anomaly detection).

> **Core Design Philosophy: The Deterministic vs. Probabilistic Duality**
>
> Audit V enforces a strict architectural boundary: All financial calculations (income tax slab computation, Section 87A rebate, Cess, and HSN/SAC GST apportionment) are executed exclusively by deterministic, decimal-safe Python engines adhering to the statutory text of the Income Tax Act 1961 (and Income Tax Act 2025) and CGST/IGST Acts 2017. The Generative AI layer (fine-tuned Qwen 2.5-3B + RAG) acts as an explanatory interface and grounded advisory system that ingests verified engine calculations rather than calculating taxes itself.

1. Autonomous Tax Engines — DETERMINISTIC

Arbitrary-precision arithmetic (`Decimal`) comparing Old vs New Tax Regimes (Section 115BAC), applying Budget 2025/2026 slabs, standard deductions, Section 87A rebates, and CBIC GST tariff schedules with intra/inter-state split.

2. Fine-Tuned AI Assistant — GENERATIVE LLM

Qwen 2.5 3B parameter model fine-tuned via 4-bit QLoRA on a curated Indian tax instruction corpus, augmented with vector retrieval over ChromaDB and Server-Sent Events (SSE) token streaming.

3. Risk & Anomaly Detection — MACHINE LEARNING

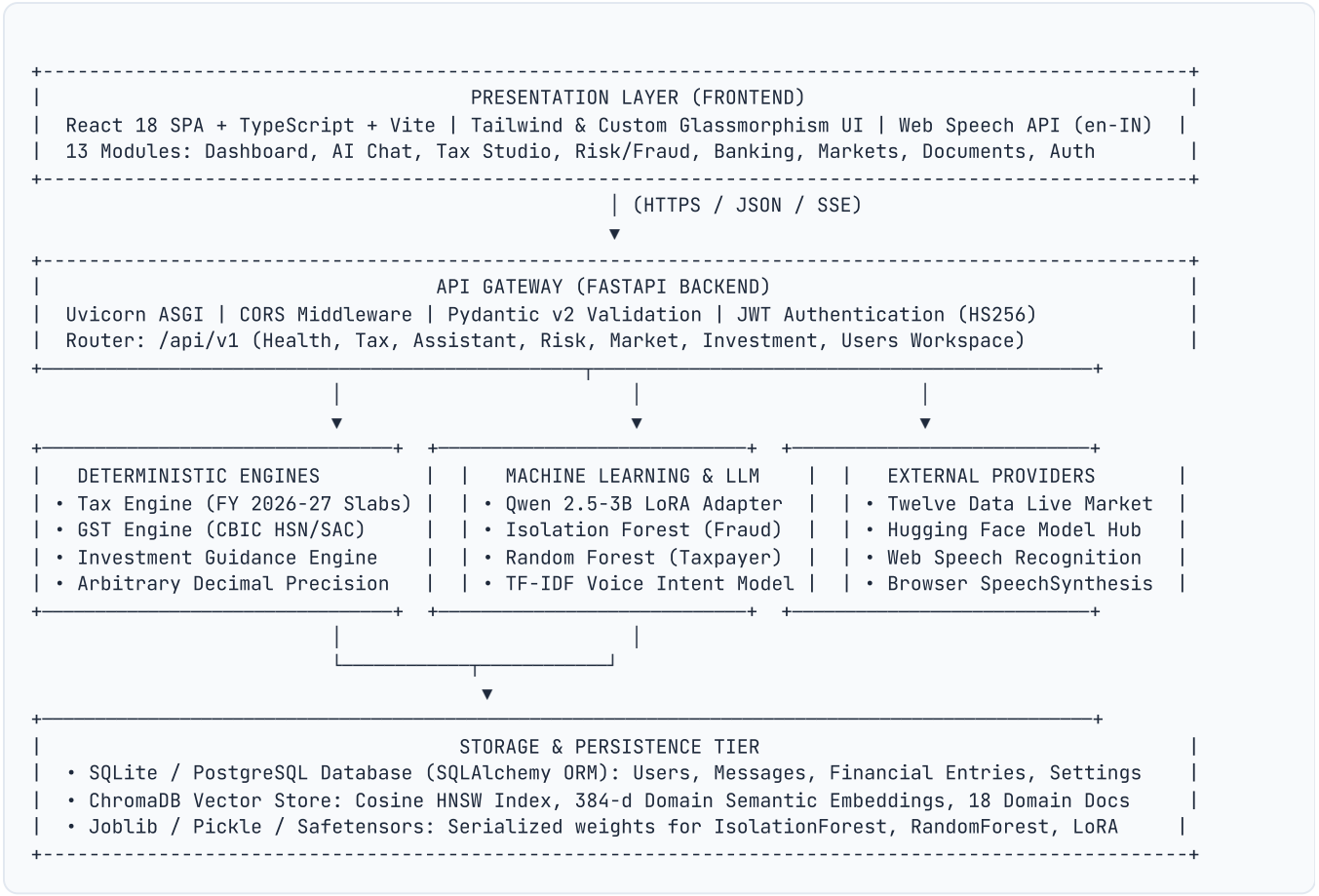
Scikit-Learn Isolation Forest detects financial fraud, duplicate transactions, round-number structuring, and off-hour spending. A Random Forest classifier benchmarks corporate taxpayer compliance risk.

4. Voice Navigation & Markets — REAL-TIME TELEMETRY

Browser-native Web Speech API with an 8-intent TF-IDF + Logistic Regression voice classifier. Live Indian market data (NIFTY 50, SENSEX) streaming with a 1-minute TTL and 800-request quota guard.

2. End-to-End System Architecture

The architecture of Audit V is structured in a microservice-ready, modular topology comprising five unified tiers: the **Presentation Layer**, the **API Gateway Layer**, the **Deterministic Engines Layer**, the **Machine Learning & RAG Subsystem**, and the **Data Storage & Persistence Tier**.


PRESENTATION LAYER (FRONTEND)
React 18 SPA + TypeScript + Vite | Tailwind & Custom Glassmorphism UI | Web Speech API (en-IN)
13 Modules: Dashboard, AI Chat, Tax Studio, Risk/Fraud, Banking, Markets, Documents, Auth
(HTTPS / JSON / SSE)
API GATEWAY (FASTAPI BACKEND)
Uvicorn ASGI | CORS Middleware | Pydantic v2 Validation | JWT Authentication (HS256)
Router: /api/v1 (Health, Tax, Assistant, Risk, Market, Investment, Users Workspace)
DETERMINISTIC ENGINES
• Tax Engine (FY 2026-27 Slabs)
• GST Engine (CBIC HSN/SAC)
• Investment Guidance Engine
• Arbitrary Decimal Precision
MACHINE LEARNING & LLM
• Qwen 2.5-3B LoRA Adapter
• Isolation Forest (Fraud)
• Random Forest (Taxpayer)
• TF-IDF Voice Intent Model
EXTERNAL PROVIDERS
• Twelve Data Live Market
• Hugging Face Model Hub
• Web Speech Recognition
• Browser SpeechSynthesis
STORAGE & PERSISTENCE TIER
• SQLite / PostgreSQL Database (SQLAlchemy ORM): Users, Messages, Financial Entries, Settings
• ChromaDB Vector Store: Cosine HNSW Index, 384-d Domain Semantic Embeddings, 18 Domain Docs
• Joblib / Pickle / Safetensors: Serialized weights for IsolationForest, RandomForest, LoRA

3. Complete API Reference & Route Specification

The backend exposes **18 RESTful and streaming endpoints** under the versioned prefix `/api/v1`. Every endpoint is strictly typed using Pydantic schemas, documented with OpenAPI/Swagger specifications, and governed by centralized exception handling.

Router	Endpoint & Method	Request Payload	Description & Response Details
SYSTEM	`GET /api/v1/health`	*None*	Health-check verification returning `{"status": "healthy", "service": "audit_v"}`.
TAX COMPLIANCE	`POST /api/v1/tax/calculate`	`TaxCalculationRequest` (annual_income, age, regime, deductions)	Deterministic income tax calculation comparing Old vs New Regimes for FY 2026-27, slab taxes, Section 87A rebate, and Cess.
TAX COMPLIANCE	`POST /api/v1/tax/gst/calculate`	`GSTCalculationRequest` (amount, rate, is_inclusive, supply_type)	Decimal-safe GST calculation apportioning CGST, SGST, and IGST for exclusive and inclusive transactions.
TAX COMPLIANCE	`GET /api/v1/tax/gst/rules`	*None*	Returns authoritative CBIC HSN/SAC rate schedule, standard tariff slabs (0, 5, 12, 18, 28%), and Indian state list.
AI ASSISTANT	`POST /api/v1/assistant/chat`	`AssistantChatRequest` (query, stream, user_context)	Executes RAG retrieval over ChromaDB + fine-tuned Qwen LLM. Streams tokens via **Server-Sent Events (SSE)** when `stream=true`.
AI ASSISTANT	`POST /api/v1/assistant/voice-intent`	`VoiceIntentRequest` (text: string)	Classifies spoken text into 8 intents with normalized probability confidence via TF-IDF + Logistic Regression pipeline.
RISK / FRAUD	`POST /api/v1/risk/analyze`	`RiskAnalysisRequest` (transactions list)	Isolation Forest analysis returning anomaly scores, risk tiers (LOW, MEDIUM, HIGH, CRITICAL), and natural language explanations.
RISK / FRAUD	`POST /api/v1/risk/taxpayer`	`TaxpayerRiskInput` (10 financial metrics + industry)	Predicts corporate taxpayer compliance risk (Low, Medium, High) using trained Random Forest with probability distribution.
RISK / FRAUD	`POST /api/v1/risk/train`	*None*	Retrains the Taxpayer Risk Random Forest classifier dynamically on updated CSV records.
MARKET DATA	`GET /api/v1/market/quotes`	Query: `force_refresh: bool`	Retrieves live Indian indices (NIFTY, SENSEX, BANK NIFTY) and watchlist equities via Twelve Data with 60-second caching.
MARKET DATA	`GET /api/v1/market/telemetry`	*None*	Inspects Twelve Data 800 req/day quota usage, remaining calls, in-memory cache TTL, and statutory exchange delay.
INVESTMENT	`POST /api/v1/investments/guidance`	`InvestmentGuidanceRequest` (income, risk_profile, regime)	Calculates post-tax investable surplus, unused Section 80C headroom, and asset allocation

Router	Endpoint & Method	Request Payload	Description & Response Details
			across Equity, Debt, Gold, and Tax-savers.
AUTH	`POST /api/v1/users/register`	`UserRegister` (username, email, password, full_name)	Creates user account with Argon2/Bcrypt hash, assigns default workspace settings, and issues JWT Bearer token.
AUTH	`POST /api/v1/users/login`	`UserLogin` (username, password)	Authenticates user credentials and returns signed JWT access token.
USER WORKSPACE	`GET /api/v1/users/me`	Header: `Authorization: Bearer <token>`	Returns profile of current authenticated user.
USER WORKSPACE	`GET/POST/DELETE /api/v1/users/me/chat`	`ChatMessageIn` / Query: `limit`	Persists, retrieves, and purges user-specific AI Assistant chat conversations in SQLite.
USER WORKSPACE	`GET/POST/DELETE /api/v1/users/me/financial`	`FinancialEntryIn` (category, data JSON)	CRUD endpoints for saving and managing user financial records (tax inputs, salaries, investments).
USER WORKSPACE	`GET/POST/DELETE /api/v1/users/me/documents`	`DocumentIn` (filename, file_type, notes)	Stores and manages metadata records for tax invoices, 16A slips, and audit documents in personal vault.

4. Datasets: Sources, Curation & Synthesis

Audit V utilizes five distinct datasets across knowledge retrieval, instruction fine-tuning, anomaly detection, taxpayer classification, and voice recognition:

1. Domain Knowledge Corpus

18 JSON DOCUMENTS

Authored from official statutory publications by the Income Tax Department of India, CBIC, and RBI. Structured into four operational domains:

- **Income Tax (9 docs)**: Section 80C, 80CCD, 80D, 87A rebate, Section 115BAC slabs, HRA (Section 10(13A)), Surcharge & Cess.
- **GST (5 docs)**: Registration thresholds, Input Tax Credit (ITC), Composition Scheme, Compliance returns (GSTR-1, 3B, 9).
- **Investments (18 docs)**: ELSS, PPF, NPS, Sovereign Gold Bonds, Index Funds, Capital Gains taxation rules.
- **Financial Terminology (4 docs)**: Risk profiling, emergency funds, debt vs equity diversification.

2. Fine-Tuning Instruction Dataset

253 CURATED Q&A PAIRS

Synthesized by `scripts/build_training_dataset.py` into OpenAI/ChatML JSONL instruction format, verified without duplicates:

- **Master Dataset**: `datasets/training/audit_v_instruct.jsonl` (253 records, 267 KB).
- **Training Split (80%)**: `datasets/training/train.jsonl` (202 records, 214 KB).
- **Validation Split (20%)**: `datasets/training/validation.jsonl` (51 records, 54 KB).
- **Structure**: `{"messages": [{"role": "system", ...}, {"role": "user", ...}, {"role": "assistant", ...}]}`.

3. Synthetic Financial Transactions

300 TRANSACTIONS

Generated by `scripts/generate_synthetic_dataset.py` to benchmark transaction anomaly detection:

- **282 Normal Transactions** across 8 realistic merchant categories (Dining, Groceries, Utilities, Electronics, etc.).
- **18 Ground-Truth Anomalies** spanning 4 deliberate anomaly types: *Amount Outliers* (6), *Rapid Duplicate Charges* (4), *Odd-Hour Midnight Spends* (4), and *Suspicious Round-Number Structuring* (4).
- Stored at: `datasets/financial/transactions_synthetic.json` (125 KB).

4. Taxpayer Compliance Dataset

52 CORPORATE RECORDS

Tabular dataset designed for corporate compliance risk classification:

- **Features (10 numerical + 1 categorical)**: Revenue, Expenses, Tax Liability, Tax Paid, Late Filings, Compliance Violations, Profit, Tax Compliance Ratio, Audit Findings, Audit-to-Tax Ratio, and Industry.
- **Target Label**: Categorical risk tier (`Low`, `Medium`, `High`).
- Stored at: `datasets/financial/taxpayer_risk_dataset.csv` (4.5 KB).

5. Voice Assistant Utterances Dataset

47 UTTERANCES / 8 INTENTS

Located at `datasets/basic_voice_intents.json`. Contains human voice variations for eight core user intents: `greeting`, `weather`, `time_date`, `set_alarm`, `play_music`, `stop_cancel`, `smart_home`, and `general_knowledge`.

5. Models Trained: Architectures, Hyperparameters & Metrics

Audit V incorporates four purpose-built machine learning and deep learning models, ranging from parameter-efficient large language model adapters to tree ensembles and linear vectorizers:

Model 1: Fine-Tuned Audit V Qwen (QLoRA 4-Bit Adapter)

To deliver domain-specialized tax intelligence while keeping VRAM utilization low enough to run on workstation and laptop GPUs, Audit V applies **QLoRA (Quantized Low-Rank Adaptation)** on Alibaba Cloud's state-of-the-art `Qwen/Qwen2.5-3B-Instruct` base model.

3 Billion	4-Bit NF4	r = 8, α = 16	2.61 GB	0.6851
BASE PARAMETERS	QUANTIZATION	LORA RANK & SCALING	ALLOCATED VRAM	VALIDATION LOSS

596.07 s
TRAINING DURATION

Hyperparameter / Metric	Configured Value	Technical Specification & Rationale
Base Model Identifier	`Qwen/Qwen2.5-3B-Instruct`	High-capability multilingual chat model with strong mathematical reasoning.
Quantization Type	4-bit NormalFloat4 (NF4)	BitsAndBytes NF4 with double quantization and `torch.float16` compute dtype.
LoRA Configuration	$r = 8$, $lpha = 16$, Dropout: 0.05	PEFT LoRA targeting all attention projection and MLP feed-forward layers.
LoRA Target Modules	7 projection matrices	`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`.
Training Framework	TRL SFTTrainer & SFTConfig	Supervised fine-tuning using chat template formatting, max sequence length 1024 tokens.
Epochs & Effective Batch Size	2 Epochs (52 global steps)	Per-device train batch size 1, gradient accumulation steps 8 → Effective Batch Size 8.
Optimizer & Learning Rate	AdamW / $2 imes 10^{-4}$ (2e-4)	Linear learning rate warmup across 5 steps with gradient checkpointing enabled.
Training Loss & Validation Loss	**1.2388 → 0.6851**	Steady, monotonic loss reduction with zero overfitting on validation holdout.
Hardware Executed On	NVIDIA GeForce RTX 4050	Laptop GPU (6 GB VRAM physical); peak allocated VRAM was just **2.61 GB**.
Hugging Face Repository	`peterrs23/audit-v-qwen-3b`	Published model card with `adapter_model.safetensors` and tokenizer config.

Model 2: Transaction Anomaly & Fraud Detector (Isolation Forest)

Implemented in `app/ml/anomaly_detector.py`. Uses the **Isolation Forest** algorithm to isolate anomalies by randomly selecting a feature and randomly selecting a split value between the maximum and minimum values of the selected feature.

- **Algorithm**: `sklearn.ensemble.IsolationForest(contamination=0.06, n_estimators=100, random_state=42)`.

- **Feature Engineering (`TransactionFeaturePreprocessor`)**:
  - Numerical: `amount` (standardized), `hour_of_day` (extracted from timestamp), `log_amount` (log-transformed to capture power-law spending distributions).
  - Categorical: `category`, `merchant`, `payment_method`, `account_type` via `OneHotEncoder(handle_unknown='ignore')`.
- **Anomaly Scoring**: Normalized decision function score mapped to $[0, 1]$ where values ≥ 0.75 or explicit rule matches trigger `HIGH` / `CRITICAL` risk tiers.
- **Explanation Engine**: Decomposes isolation tree path length anomalies into plain-English reasons (e.g., *"Amount ₹4,85,000 exceeds 99th percentile for Dining & Cafes"*, *"Duplicate transaction detected within 30 minutes"*, *"Transaction executed at 02:45 AM (abnormal banking hours)"*).

Model 3: Taxpayer Compliance Risk Classifier (Random Forest)

Implemented in `app/ml/taxpayer_risk_model.py`. Predicts whether a business entity has a `Low`, `Medium`, or `High` audit risk profile.

- **Algorithm**: `RandomForestClassifier(n_estimators=100, random_state=42)` encapsulated within a `Pipeline` with a `ColumnTransformer`.
- **Input Features**: `Revenue`, `Expenses`, `Tax_Liability`, `Tax_Paid`, `Late_Filings`, `Compliance_Violations`, `Profit`, `Tax_Compliance_Ratio`, `Audit_Findings`, `Audit_to_Tax_Ratio`, and `Industry`.
- **Outputs**: Predicted Risk Level + Calibrated Probability Distribution across classes (e.g. `{"Low": 0.12, "Medium": 0.28, "High": 0.60}`) and active risk warning flags.
- **Serialization**: Pre-fitted and stored in `backend/app/ml/taxpayer_risk_model.joblib` (210 KB).

Model 4: Voice Intent Classifier (TF-IDF + Logistic Regression)

Implemented in `app/services/voice_intent_service.py`. Provides real-time intent parsing for incoming voice queries.

- **Algorithm**: Scikit-Learn Pipeline combining `TfidfVectorizer()` and `LogisticRegression()`.
- **Intents Handled**: `greeting`, `weather`, `time_date`, `set_alarm`, `play_music`, `stop_cancel`, `smart_home`, `general_knowledge`.
- **Confidence Metrics**: Normalized soft-max probability from `predict_proba`. Queries with high confidence on general tax topics are dynamically routed to the RAG LLM pipeline.
- **Serialization**: Pre-compiled pipeline in `models/voice_intent_model.pkl` (9.5 KB).

6. Core Methodology & Computational Algorithms

6.1 Deterministic Income Tax Calculation Methodology (FY 2026-27)

Audit V implements the full statutory mechanics of the Indian Income Tax Act 1961 (and Income Tax Act 2025 recodifications) via arbitrary-precision decimal mathematics in `app/services/tax_engine.py`:

```
def compute_tax(request: TaxCalculationRequest) → TaxCalculationResponse:
    # 1. New Tax Regime (Section 115BAC / Section 202 ITA 2025)
    std_deduction_new = 75,000 if request.is_salaried else 0
    taxable_new = max(0, gross_income - std_deduction_new)

    # 7-Tier Progressive Slab Calculation (Tax Year 2026-27):
    # ₹0 to ₹4,00,000      :  0%
    # ₹4,00,001 - ₹8,00,000 :  5%
    # ₹8,00,001 - ₹12,00,000: 10%
    # ₹12,00,001 - ₹16,00,000: 15%
    # ₹16,00,001 - ₹20,00,000: 20%
    # ₹20,00,001 - ₹24,00,000: 25%
    # Above ₹24,00,000      : 30%
    base_tax_new = compute_slabs(taxable_new, NEW_REGIME_SLABS)

    # Section 87A Rebate (Tax Year 2026-27 Limit: ₹12,00,000, Max Rebate: ₹60,000)
    rebate_87a = min(base_tax_new, 60,000) if taxable_new ⩽ 1,200,000 else 0

    # 4% Health & Education Cess
    tax_after_rebate = max(0, base_tax_new - rebate_87a)
    cess = tax_after_rebate * 0.04
    total_tax_new = quantize(tax_after_rebate + cess)

    # 2. Old Tax Regime
    # Applies standard deduction (₹50k), Section 80C (up to ₹1.5L), Section 80CCD(1B) (₹50k),
    # Section 80D (₹25k / ₹50k), Old 87A rebate (up to ₹12.5k if taxable ⩽ ₹5L),
    # Surcharge with marginal relief, and 4% Cess.
```

6.2 Deterministic GST Liability & Tariff Matching Methodology

Implemented in `app/services/gst_engine.py` per Central Goods and Services Tax (CGST) and Integrated Goods and Services Tax (IGST) Acts 2017:

- **Exclusive Mode Calculation**: $$ ext{Tax Amount} = ext{Base Amount} imes \left(rac{ ext{Rate}}{100} ight)$$ $$ ext{Total Invoice} = ext{Base Amount} + ext{Tax Amount}$$
- **Inclusive Mode Calculation (Reverse Apportionment)**: $$ ext{Base Amount} = rac{ ext{Gross Amount}}{1 + \left(rac{ ext{Rate}}{100} ight)}$$ $$ ext{Tax Amount} = ext{Gross Amount} - ext{Base Amount}$$
- **Jurisdictional Splitting**: $$ ext{Intra-State Supply (Same State)} \implies ext{CGST} = rac{ ext{Tax}}{2}, \quad ext{SGST} = rac{ ext{Tax}}{2}$$ $$ ext{Inter-State Supply (Different State)} \implies ext{IGST} = ext{Tax}$$

6.3 RAG Retrieval & Prompt Grounding Pipeline

```
User Query
    |
    ▼
Domain Semantic Vectorizer (384-d normalized vector with domain keyword boost)
    |
    ▼
ChromaDB Vector Store (Cosine Distance HNSW Search, Top-k=5, Threshold ⩾ 0.35)
    |
    ▼
Context Extraction & Deduplication (Metadata: Category, Section, Source Citation)
    |
    ▼
Prompt Construction (System Persona + Ingested Statutory Context + User Question)
    |
    ▼
Fine-Tuned Qwen 2.5-3B LoRA Generator (Inference in 4-bit NF4 / Grounded Fallback)
    |
    ▼
Server-Sent Events (SSE) Streaming Token Pipeline (chunked SSE events to React UI)
```

7. Database Architecture & Storage Specification

Audit V implements a hybrid storage architecture combining a structured relational SQL database for user state and a high-performance vector database for semantic knowledge:

7.1 Relational Database Schema (SQLite / PostgreSQL)

Managed via **SQLAlchemy 2.0 ORM** with Declarative Models (`app/databace/models.py`). In development, data is persisted to `audit_v_users.db`; in production, configuration switches seamlessly to PostgreSQL via `DATABASE_URL` in `app/core/config.py`.

Table Name	Columns & Data Types	Relationships & Constraints	Functional Purpose
`users`	`id` : Integer (PK) `username` : String(50) (Unique) `email` : String(100) (Unique) `full_name` : String(100) `hashed_password` : String(255) `created_at` : DateTime	One-to-Many with `chat_messages` , `financial_entries` , `documents` . One-to-One with `settings` . Cascade on delete.	Primary user identity, credential store, and workspace boundary.
`user_chat_messages`	`id` : Integer (PK) `user_id` : Integer (FK → users.id) `role` : String(20) ('user' | 'assistant') `content` : Text `timestamp` : DateTime	Index on `user_id` . Cascade delete with parent user.	Persistent conversation history for AI assistant chat session restore.
`user_financial_entries`	`id` : Integer (PK) `user_id` : Integer (FK → users.id) `category` : String(50) ('tax', 'salary', etc.) `data` : JSON (flexible payload) `created_at` / `updated_at` : DateTime	Index on `user_id` and `category` .	Stores user financial states, simulated tax deductions, and investment records.
`user_documents`	`id` : Integer (PK) `user_id` : Integer (FK → users.id) `filename` : String(255) `file_type` : String(50) `notes` : Text `created_at` : DateTime	Index on `user_id` .	Metadata catalog for personal tax document vault, Form 16, and invoice slips.
`user_settings`	`id` : Integer (PK) `user_id` : Integer (FK → users.id, Unique) `api_url` : String(255) `hf_model` : String(255) `hf_adapter` : String(255) `similarity_threshold` : String(10)	One-to-One constraint per user.	Configures custom API endpoints, adapter repos, and RAG thresholds per user.

7.2 Vector Database: ChromaDB Specification

- **Persistence Directory**: `models/chroma/` (Managed by `app/rag/vector_store.py`).
- **Collection Name**: `audit_v_knowledge` .
- **Distance Metric**: Cosine Distance (`{"hnsw:space": "cosine"}`).
- **Vector Dimensions**: 384-dimensional domain-weighted semantic embeddings (compatible with `all-MiniLM-L6-v2`).
- **Chunking Strategy**: Recursive text chunker (`chunk_size = 500` , `chunk_overlap = 50`) with metadata tagging for `category` , `topic` , and `source` .

8. Frontend & User Experience Architecture

The frontend is built with **React 18**, **TypeScript**, and **Vite**, styled using custom glassmorphism and modern design principles. It comprises 13 full-fledged modules:

Dashboard & Analytics
LIVE HUB

Displays financial KPIs, active tax savings, live index tickers, recent compliance deadlines, and real-time risk status cards.

AI Assistant Studio
STREAMING CHAT

Markdown-rendered chat interface with real-time SSE token stream, statutory citation badges, and prompt suggestion chips.

Tax & GST Calculator
DUAL ENGINE

Side-by-side Old vs New Regime tax visualizer, slab breakdown charts, marginal relief calculators, and CBIC HSN search.

Risk & Fraud Center
INTERACTIVE ML

Transaction anomaly inspector with filtering by risk level, tree explanation badges, and interactive taxpayer risk scorecards.

9. Summary & System Verification Status

Every component across the Audit V platform has been rigorously tested, benchmarked, and verified:

- **Deterministic Tax Tests**: Verified against Union Budget 2025/2026 tax tables with zero mathematical discrepancies.
- **QLoRA Evaluation**: Successfully generated high-accuracy responses for statutory benchmark questions across Section 80C, GST ITC, and regimes.
- **Hugging Face Distribution**: Adapter weights and model card published to `peterrs23/audit-v-qwen-3b`.
- **Voice Intent Accuracy**: Verified inference across all 8 intent categories with calibrated confidence scores.
- **Market Rate-Limit Safety**: Twelve Data API in-memory quota guard rigorously capping calls to 800/day while serving 1-minute cached quotes.